SOC L1 LAB · INC-005

Detection, Mitigation, and Triage Guide: INC-005

A scan is a question, not a verdict

INCIDENT	INC-005, Outbound port scan
REPORT DATE	2026-11-01
CLASSIFICATION	Lab exercise, sanitized for publication
ENVIRONMENT	Wazuh 4.14.7 · Debian 12 · QEMU/KVM
SEVERITY	Medium

Independent portfolio lab. No production systems, client data, or real credentials.

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Audience: an L1 analyst who receives the alert, and an L2 or detection engineer who owns the network rule.

This guide uses INC-005 as the worked example. Wazuh rule 86601 is stock. The Suricata rule is local.

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1. Detection model

A port scan is a network event, but the decision needs three layers.

Layer	Source	Question	INC-005 evidence
Network	Suricata	Did a host sweep ports?	sid 1000001, 144 alerts
Host	auditd / FIM	Did the host do anything else?	none
Identity	sshd / sudo	Did someone log in or elevate?	none

If only the network layer fires, the finding is reconnaissance. The other layers decide whether it is a compromised host or an approved scanner.

1.1 Signal hierarchy

```
Low     one connection to one port
        ----  normal traffic

Medium  many SYNs in a short window
        ----  scan candidate; verify the source

High    scan + authentication failures / exploit attempts
        ----  likely compromised host

Critical scan + successful session / payload / lateral movement
```

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2. Rule analysis

2.1 The local rule

```
alert tcp $HOME_NET any -> $EXTERNAL_NET any (
  msg:"LAB SCAN outbound SYN sweep";
  flags:S;
  detection_filter: track by_src, count 50, seconds 5;
  classtype:attempted-recon;
  sid:1000001; rev:1;
)
```

- **Direction**
  `$HOME_NET -> $EXTERNAL_NET`. This is an **outbound** scan. ET Open's scan signatures are largely inbound (`$EXTERNAL_NET -> $HOME_NET`), so they missed the lab's outbound case.
- `flags:S`: counts SYN packets only, not established data.
- `detection_filter`: 50 SYNs from the same source in 5 seconds.
- classtype: `attempted-recon` (severity 2).

2.2 Why not rely on ET SCAN

ET Open has good scan signatures, but they are direction-sensitive and often target known tools/patterns. The outbound case needed an explicit rule. A production estate should use both: ET SCAN for inbound/known tools, a local rule for outbound discovery.

2.3 False-positive boundary

Traffic	Alerts
Normal `apt` + `curl`	0
40 requests in a burst	0
60 requests in a burst	10

The rule counts packets, not unique destinations. A busy client or load test crosses the threshold. Tuning options:

- raise the count (e.g., 100/5 s) and accept slower detection.
- baseline known scanners and load generators.
- correlate with authentication failures or process activity.
- use a flow-based detector for unique-destination counting.

2.4 Gaps

Gap	Consequence
Outbound-only	An inbound scan of the VM is not covered
Packet-count threshold	Legit bursts false-positive. Slow scans evade
One destination	A subnet sweep is not represented
No follow-on correlation	Scan + exploit are separate alerts

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3. Triage playbook

3.1 Five questions

1. Who is the source? Internal host, scanner, jump host, or unknown?
2. What was scanned? One host, a subnet, specific ports?
3. Is it approved? Scanner schedule, change ticket, asset inventory.

4. What followed? Auth failures, exploit attempts, new sessions, egress.
5. Is the source healthy? Process/file anomalies, unusual logins, malware alerts.

3.2 Decision tree

```
Port scan alert
│
├─ Source is a known scanner and the window is scheduled
│   └─ Close. Verify scope and rate
│
├─ Source is an internal host, no approval
│   ├─ No follow-on activity: Medium. Contact the owner
│   ├─ Auth failures / exploit attempts: High. Isolate
│   └─ Session / payload / lateral movement: Critical
│
└─ Source is external
    ├─ Blocked by edge: informational
    └─ Reached an internal host: High. Check exposure
```

3.3 Step by step

1. Confirm the direction: outbound or inbound.
2. Identify the source and destination from the alert.
3. Check the scanner inventory and change calendar.
4. Pivot to host telemetry for the source: logins, `sudo`, new processes, FIM changes.
5. Pivot to authentication logs for the destination.
6. Check for egress or a second stage.
7. Assign severity with §4.
8. Contain only if there is evidence of compromise. A scan alone does not justify isolating a production scanner.

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4. Severity matrix

Situation	Severity
Approved scanner, scheduled window	Low / close
Internal host, no approval, no follow-on	Medium
Internal host + authentication failures	High
Internal host + exploit attempts	High
Scan + successful session / payload	Critical
External scan blocked at the edge	Low
External scan reaching an internal service	High

INC-005: internal host, no approval, no follow-on to Medium.

5. Containment and mitigation

5.1 During the incident

Action	When	Caution
Contact the host owner	Medium, no follow-on	Fastest path to "approved scanner"
Capture the flow/alert set	Always	Before rotating logs
Isolate the host	Only with compromise evidence	A scanner may be production
Block the source	External/edge only	Internal blocking can break monitoring
Check the destination	Always	Open ports may be the real finding

5.2 Hardening

- Maintain a scanner inventory with source addresses and schedules.
- Segment management/scanner traffic from user traffic.
- Restrict egress from the web tier.
- Alert on scan to auth-failure sequences.
- Add the inbound rule if the sensor sees inbound traffic.
- Keep the IDS feed and rule thresholds reviewed.

5.3 Detection controls

Control	Purpose
Local outbound rule	Catches the direction ET misses
ET SCAN rules	Known tools and inbound scans
5710/5760/5763	Follow-on authentication attempts
80792 / FIM	Host activity after the scan
86601	Wazuh ingestion of Suricata alerts

6. Escalation template

```
INC-005: Outbound port scan

What:  10.0.2.15 scanned 200 TCP ports on 10.0.2.2 at 23:26:35 UTC. 144 Suricata
       alerts, signature LAB SCAN outbound SYN sweep (sid 1000001).
Scope: Source is the lab VM; destination is the gateway. One host, one target.
Follow: No authentication, file change, process execution, or egress observed.
Done:  Evidence preserved; no containment (recon only).
```

```
Ask:   (1) Was a scan scheduled for this host? (2) If not, review host telemetry for
       compromise. (3) Should the threshold be raised for the load-test window?
```

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7. Post-incident

- Update the scanner inventory or the threshold.
- Add the scan + auth-failure correlation test.
- If the bridge upgrade is deployed, add the inbound case.
- Review whether the 50/5 s threshold is right for the estate.
- Document the direction limitation in the runbook.

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8. Common mistakes

Mistake	Consequence
Treating a scan as a breach	Wasted isolation, angry scanner owner
Closing because "it is just recon"	Misses a compromised host doing discovery
Relying only on ET SCAN	Outbound scans are missed
Ignoring the destination	The open port may be the real issue
Raising the threshold without a baseline	Slow scans evade. Load tests still fire
Blocking an internal scanner	Breaks security operations

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9. Coverage map

Technique	ID	Covered by	Gap
Network service discovery	T1046	sid 1000001, ET SCAN	Outbound-only local rule
Active scanning	T1595	sid 1000001	Slow/distributed scans
Follow-on auth	T1110	5710/5760/5763	Needs correlation
Exploitation	T1190	Suricata web rules	Not in this scenario

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10. Metrics

Metric	INC-005
Normal traffic alerts	0
40-request burst	0
60-request burst	10 (FP)
Scan alerts	144
Time from scan to Wazuh	~1 s
Follow-on activity	none
Severity	Medium

The detection is fast. The judgement is whether the source is allowed to scan.